

Lec21

Why jquery is created?

Before jQuery, JavaScript DOM manipulation was lengthy and difficult.

Example in JavaScript:

```
document.getElementById("title").style.color = "red";
```

Same in jQuery:

```
$("#title").css("color", "red");
```

Shorter, easier, cleaner.

Introduction to jQuery

Query is a fast and lightweight JavaScript library.

It simplifies:

- HTML DOM manipulation
- Event handling
- Animations
- AJAX

Officially, jQuery describes it as “write less, do more.”

Why jQuery?

Without jQuery:

- More code
- Complex syntax
- Browser compatibility issues

With jQuery:

- Less code
- Easier syntax
- Cross-browser support

Example

JavaScript:

```
document.getElementById("btn").addEventListener("click", function(){  
    alert("Clicked");  
});
```

jQuery:

```
$("#btn").click(function(){  
    alert("Clicked");  
});
```

How to Add jQuery

Use CDN.

```
<script src="https://code.jquery.com/jquery-4.0.0.min.js"></script>
```

Or download from [Official jQuery Download Page](#)

jQuery Syntax

Basic syntax:

```
$(selector).action();
```

Breakdown:

- `$` → jQuery
- `selector` → select HTML element
- `action()` → perform operation

Example:

```
$("#p").hide();
```

Meaning:

- Select all paragraphs
- Hide them

Document Ready

Run code only after HTML fully loads.

```
$(document).ready(function(){  
    // code here  
});
```

Shortcut:

```
$(function(){  
    // code here  
});
```

Example:

```
$(document).ready(function(){  
    $("p").hide();  
});
```

jQuery Selectors

Selectors are used to find HTML elements.

Example HTML:

```
<h1 id="heading">Title</h1>
```

```
<p class="para">Paragraph 1</p>
```

```
<p class="para">Paragraph 2</p>
```

ID Selector

Use #

```
$("#heading")
```

Example:

```
$("#heading").hide();
```

Class Selector

Use .

```
$(".para")
```

Example:

```
$(".para").hide();
```

Element Selector

`$("p")`

Selects all <p> tags.

Universal Selector

`$("*")`

Selects all elements.

Multiple Selectors

`$("h1, p")`

Selects both.

First / Last Selectors

First:

```
$("p:first")
```

Last:

```
$("p:last")
```

Even / Odd

```
$("tr:even")
```

```
$("tr:odd")
```

Events in jQuery

Events = user actions.

Examples:

- click
- dblclick
- mouseenter
- mouseleave
- keypress

Common Event Methods

click()

```
$("#btn").click(function(){  
    alert("Button clicked");  
});
```

dblclick()

```
$("#btn").dblclick(function(){  
    alert("Double clicked");  
});
```

mouseenter()

```
$("#box").mouseenter(function(){  
    console.log("Mouse entered");  
});
```

mouseleave()

```
$("#box").mouseleave(function(){  
    console.log("Mouse left");  
});
```

keypress()

```
$("#input").keypress(function(){  
    console.log("Typing");  
});
```

focus()

```
$("#input").focus(function(){  
    console.log("Focused");  
});
```

blur()

```
$("#input").blur(function(){  
    console.log("Left input");  
});
```

on() Method

Used for multiple events or dynamic elements.

```
$("#btn").on("click", function(){  
    alert("Clicked");  
});
```

Multiple events:

```
$("#btn").on({  
    click: function(){  
        console.log("Clicked");  
    },  
    mouseenter: function(){  
        console.log("Hover");  
    }  
});
```

jQuery DOM Manipulation

Content Methods

`text()`

Gets or sets text.

```
$("#heading").text("New Title");
```

html()

Gets or sets HTML.

```
$("#heading").html("<i>New Title</i>");
```

val()

Works with forms.

```
$("#input").val();
```

Set value:

```
$("#input").val("Hello");
```

Attribute Methods

attr()

Get attribute:

```
$("#img").attr("src");
```

Set attribute:

```
$("#img").attr("src", "new.jpg");
```

CSS Manipulation

css()

```
$("#heading").css("color", "red");
```

Multiple:

```
$("#heading").css({  
  color: "red",  
  fontSize: "30px"  
});
```

Add/Remove Classes

addClass()

```
$("#box").addClass("active");
```

removeClass()

```
$("#box").removeClass("active");
```

toggleClass()

```
$("#box").toggleClass("active");
```

DOM Element Manipulation

append()

Add at end.

```
$("#box").append("<p>New Paragraph</p>");
```

prepend()

Add at beginning.

```
$("#box").prepend("<p>New Paragraph</p>");
```

after()

Add after element.

```
$("#box").after("<p>After Div</p>");
```

before()

Add before element.

```
$("#box").before("<p>Before Div</p>");
```

remove()

Delete element.

```
$("#box").remove();
```

jQuery Animations

hide() / show()

Hide:

```
$("#box").hide();
```

Show:

```
$("#box").show();
```

With speed:

```
$("#box").hide(1000);
```

Fade Effects

fadeIn()

```
$("#box").fadeIn();
```

fadeOut()

```
$("#box").fadeOut();
```

fadeToggle()

```
$("#box").fadeToggle();
```

fadeTo()

```
$("#box").fadeTo("slow", 0.5);
```

Side Effects

slideDown()

```
$("#panel").slideDown();
```

slideUp()

```
$("#panel").slideUp();
```

slideToggle()

```
$("#panel").slideToggle();
```

animate()

Custom animation.

Example:

```
$("#box").animate({  
  left: "250px",  
  opacity: "0.5"  
});
```

Chaining

Query allows multiple methods in one line.

```
$("#box")  
  .css("color", "red")  
  .slideUp(1000)  
  .slideDown(1000);
```

Very powerful.

Example

HTML:

```
<button id="btn">Click</button>  
<div id="box">Hello</div>
```

jQuery:

```
$(document).ready(function(){  
    $("#btn").click(function(){  
        $("#box").slideToggle();  
    });  
});
```

Click button → box hides/shows.

Javascript vs jQuery

JavaScript	jQuery
More code	Less code
Complex syntax	Easy syntax
Manual DOM handling	Simplified DOM handling

Task

Rebuild a DOM feature using jQuery